

Lesson 8 — Forever Loops & Controlled Stopping

Matches LEGO Teacher Portal lesson: "Wake-Up Call".

Objective (I can...): I can use a forever loop and stop it in a controlled way with an event or a condition.

CS concept: Infinite loops and controlled termination · **Builds on:** Lesson 7 (loop bodies).

AI Connection (2 min): An always-on assistant runs a forever loop, sampling the microphone until a wake word fires. *Impacts prompt:* "An always-listening device is convenient — what's the tradeoff for the people living with it?"

Materials: LEGO CS & AI kit + device per pair, notebooks. **Model reuse:** Loops model from Lesson 6 — code only.

60-minute flow - Warm-up (8): "What stops it?" — coach starts an endless chant; pairs propose a clean stopping rule (a signal, not just "give up"). Introduce controlled termination. - **Concept + demo (10):** Demo a forever loop (blinking light), stop it with the Stop button, then better, with a button event — contrast an abrupt kill with a designed exit. - **Build/Code (30):** Pairs program a forever loop (constant patrol or blink) on the built model, then engineer a controlled stop via an event or a sensor condition. They note in the notebook the difference between the two stopping methods. *Swap at 15 min.* -

Challenge (7): Use one event to start the forever loop and a different event to stop it, so the model has a clear on/off life cycle. - **Share + pack-up (5):** Pairs demo start and controlled stop and explain why a designed exit beats the Stop button. Model stays built.

Pair roles: Driver codes the loop, Navigator owns the stop design. Swap at 15 min. (*Trio rotates start/loop/stop.*)

Differentiation: *Support* — stop via the Stop button, then add one event stop with help. *Extension / fast-finisher* — add a condition that also halts the loop on a sensor threshold. *Reteach* — run the forever loop, feel that it never ends, then add the stop together.

Exit check: Pair runs a forever loop and halts it through a designed event or condition, not just Stop. **At-home (optional):** Name a system that "runs until told to stop" (a thermostat, a server); describe its stop condition. ~10 min.
